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Assessing Sustainability Efforts and Greenwashing in Global Oil and Gas Supply Chain- Natural Language Processing Approach

Sanchari Ghosh
Department of Management Studies
and Industrial Engineering
Indian Institute of Technology (ISM),
Dhanbad
Jharkhand, India
22dr0240@iitism.ac.in

Sandeep Mondal
Department of Management Studies
and Industrial Engineering
Indian Institute of Technology (ISM),
Dhanbad
Jharkhand, India
sandeepmondal@iitism.ac.in

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***Abstract*— Global oil and gas supply chain ensures energy security of the world; however, faces significant compliance pressure on environmental footprints and allegation of overwhelming in sustainability reporting from the society. This paper proposes a natural language processing based unified framework to evaluate the environmental compliances as per global standards and greenwashing detection. It extracts keywords and numeric data from the sustainability reports, analyses sentiments for the entire supply chain operations, and matches with the pre trained global standard thresholds as per the United Nation Climate Action Declaration 2022 to classify the compliances and mark the probable greenwashing. Thus, the study offers a valuable tool for stakeholders and regulators to assess accuracy and transparency in corporate sustainability practices.**

Keywords*— *Sustainability Compliance, Greenwashing Detection, Natural Language Processing

I. Introduction

The global oil and gas supply chain (OGSC) plays a vital role in fuelling the world economy yet faces intense scrutiny due to its environmental impacts, with the increasing concerns about climate change and natural resources depletion, various governments worldwide have now mandated the publication of environment social and governance (ESG) reports periodically. Thus, the companies are increasingly compelled to publish the ESG reports annually to showcase their commitment towards sustainability. However, scepticisms are rising on the transparency and accuracy of the reports with the allegations of greenwashing where companies exaggerate and misrepresent their ESG efforts [1]-[3].

Scrutinising the ESG reports to assess the credibility and corporate adherence towards sustainability goals poses significant challenges. Traditional analysis might be subjective as well as labour intensive too, thus creates a need of efficient scalable time bound and objective solution. Machine learning (ML) stands here as a promising alternative by automating the evaluation process of such extensive data into meaningful summaries with limited subjectivity and least time [4].

ML techniques such as natural language processing (NLP) has emerged as popular method for automating the analysis of large unstructured text data expediently. NLP is able to facilitate in extracting essential metrics in the reports, recognising patterns in corporate language, and thus highlights discrepancies between stated claims and actual performance. Techniques such as tokenization, named entity recognition (NER), sentiment analysis, etc. enable the processing of large volumes of text, reveal subtle signs of greenwashing and evaluate the credibility of sustainability commitments [4]-[6].

The objective of this paper is to build a framework that leverages NLP to assess the sustainability efforts for OGSC by evaluating environmental compliances against established benchmarks proposed under United Nation Sustainability Development Goals (UNSDGs) and detect greenwashing as per UN Climate Action Declaration 2022.

Next sections elaborate related works, followed by methodology, implementation framework, data analysis and result, and conclusion, respectively.

II. Related Works

Researchers have developed models to monitor corporate adherence to environmental regulations and standards, including compliance with the UNSDGs. These models include unsupervised learning frameworks that automatically classify companies based on their compliance with environmental regulations by extracting features from sustainability reports [7]. A study on applied deep learning develops a model to assess the environmental compliance of companies operating in the energy sector [8]. However, their work did not focus on unstructured textual data from sustainability reports, which can provide deeper insights into corporate intentions and practices. Greenwashing detection in sustainability reports has recently become a rising concern as more organizations are engaging in sustainability reporting practices [1]-[2]. Models build on applying NLP and deep learning techniques leverages sector-specific compliance indicators and ESGs [9]. Sentiment analysis is a method used in assessing the tone of reports and advertisements and detecting over-optimistic or neutral languages [10]. Combining sentiment analysis with sustainability indicators provides a more comprehensive assessment on such company's claims. Machine learning models trained on labelled data have proven promising in detecting greenwashing [11]. Previous studies suggest for creating benchmarks on greenwashing detection [12]. Some ML models show unsupervised learning methods on unstructured

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ESG data for better generalisation across industries [13]-[15]. Another set of studies explores the usages of ML algorithms in optimising supply chain operations while ensuring environmental sustainability. Such work also emphasizes how the ML models could predict supply chain disruptions and also recommends sustainable operation strategies [16]-[17].

Thus, existing literatures lack in a targeted focus on using NLP in assessing compliances across this oil and gas sector. So, this study tries to develop a framework to evaluate sustainability compliance and greenwashing detection specifically for the OGSC.

III. Methodology

The proposed framework has been designed to process reports by extracting certain pre-defined features related to environmental compliances and greenwashing. OGSC broadly distributed in three segments: extraction of crude oil from the borewells, i.e., Upstream segment; storage and transportation of crude to refineries under Midstream segment; and finally refining crude into final products and distributions in consumer market, i.e., Downstream segment [18]. To assess the complete chain, 5 leading companies under each stream, i.e., a total 15 companies are selected whose ESG reports for the year 2023 are considered for further analysis.

A. Feature Selection

Feature Selection is performed to reduce the input, ignore the noise and pick only relevant datasets for the analysis. Sustainability efforts will be different under various supply chain operations. Greenwashing actions can be traced by figuring out statements like claiming to be on track to reduce emissions, being vague about SDG target achievements, using misleading labels, emphasising on environmental attributes, or communicating sustainability attributes in isolation. Table 1 demonstrates such features on the OGSC sustainability efforts and greenwashing detection stream-wise.

Table 1: Feature Selection under Different Supply Chain Stages for Sustainability Efforts and Greenwashing Detection

Supply Chain Stage	Feature Selection for Sustainability Efforts	Feature Selection for Greenwashing Detection
Upstream	Emission reduction targets, such as green-house gas (GHG) emissions, flaring reductions (UNSDG 13).	Vague or unsupported claims on emission reductions which lack supporting data.
	Water usage management during crude extractions (UNSDG 6).	Overly positive language on energy efficiency without any supporting data and/or methods.
Midstream	Pipeline safety and leakage prevention (UNSDG 12).	Exaggerated claims on low emissions in transportation without quantitative measures.
	Energy efficient transportation (UNSDG 7).	Renewable energy usage claims without specific metrics.
Downstream	Emission and waste management, refinery onwards (UNSDG 13).	Greenwashing in product labelling or marketing without metrics.
	Renewable energy usages in refining (UNSDG 7).	Misleading language on energy mix and distributions.

B. System Design

The system is designed under 5 different modules to ensure scalability, readability and flexibility of the study. Details are shown in Table 2.

Table 2: Module Details for the Proposed Framework

Module	Module Name	Activities
1	Data Extraction Module	Raw text data has been extracted from the ESG reports with .pdf extension, while ensuring text accuracy from various formats.
2	Text Preprocessing Module	Extracted texts are cleaned and pre-processed like tokenization, stop-word removal, lemmatization, thus prepares for standardised text analysis.
3	Sentiment Analysis Module	Tone of the sustainability claims are analysed as positive, neutral, or negative by using a pre-trained sentiment model.
4	Environmental Compliance Module	An environmental compliance score is calculated for each company based on the extracted sustainability metrics (stream-wise) benchmarked with relevant UNSDGs and labelled as compliant, partially compliant or non-complaint based on score.
5	Greenwashing Detection Module	Green-washings are identified by flagging exaggerated and/or unsupported sustainability claims using keyword matching and sentiment analysis.

C. Sequence Diagram

Figure 1 illustrates the interactions among different modules in a given sequence, delineated by arrowheads.

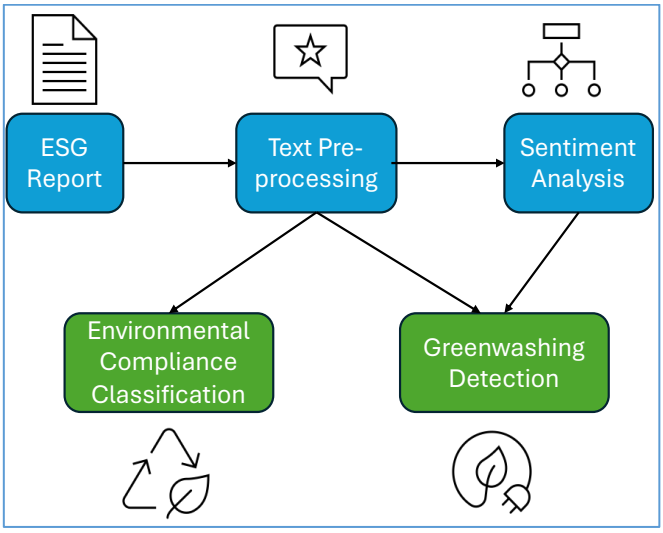

Figure 1: Sequence Diagram of the Model

The system processes ESG reports as text-based files (.pdf), cleans unnecessary words and tokenizes relevant texts, accordingly, analyses the sentiments, evaluates sustainability compliances and finally detects greenwashing.

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IV. Implementation Framework

The following algorithms are used to analysis the framework.

A. Algorithm 1: Data Extraction and Preprocessing

It extracts textual data from ESG reports and prepares the texts according to module 1 and 2, respectively.

Input: ESG reports in .pdf format

1. Load the ESG report using PDF parsing tools.
2. Extract raw text from the report.
3. Preprocess the text:
   a. Convert text to lowercase.
   b. Remove special characters and digits using regular expressions.
   c. Tokenise the text into individual words.
   d. Remove stop-words from the tokenized text.
   e. Apply lemmatization to reduce words to their base form.
4. Store the cleaned and pre-processed text for analysis.

Output: Pre-processed text data ready for NLP analysis.

B. Algorithm 2: Sentiment Analysis

This algorithm performs sentiment analysis for tone identification of sustainability claims and flag potential greenwashing statements.

Input: Pre-processed text data from ESG report

1. Load a pre-trained sentiment analysis model.
2. For each sustainability-related claim in the text:
   a. Analyse the sentiment of the text
   b. Classify the sentiment
3. Flag potential greenwashing:
   a. If sentiment is overly positive without supporting data, flag as "Possible Greenwashing."
   b. If the sentiment is neutral or negative with evidence, classify as "Legitimate Claim."
4. Store sentiment classifications and greenwashing flags.

Output: Sentiment classifications and flagged greenwashing instances.

C. Algorithm 3: Sustainability Assessment (Environmental Compliance)

These algorithms assess sustainability efforts across various segments of OGSC.

a) Algorithm 3.1: Upstream Sustainability Assessment (Environmental Compliance)

Input: Cleaned text data from ESG report for upstream

1. Extract sustainability metrics relevant to upstream operations:
   a. Emissions data (UNSDG 13)
   b. Water usage (UNSDG 6)
2. Compare extracted metrics with predefined thresholds for compliance:
   a. Emissions reduction > X% → Compliant
   b. Water usage reduction targets > Y% → Compliant
   c. If reductions are between X% and (X-Δ) % for emissions or Y% and (Y-Δ) % for water usage → Partially compliant
   d. Else → Non-compliant
3. Classify environmental compliance based on results.

Output: Sustainability compliance classification (Compliant, Non-compliant, Partially compliant)

b) Algorithm 3.2:Midstream Sustainability Assessment

Input: Cleaned text data from ESG reports

1. Extract sustainability metrics relevant to the midstream operations:
   a. Energy efficiency in transportation (aligned UNSDG 7)
   b. Pipeline safety and leak prevention (UNSDG 12)
2. Compare extracted metrics with predefined thresholds:
   a. Energy efficiency improvement > X% → Compliant
   b. Pipeline safety incident <Y→ Compliant
   c. If improvements are between X% and (X-Δ) % for energy efficiency or safety incidents are only slightly above Y% → Partially compliant
   d. Else → Non-compliant
3. Classify environmental compliance.

Output: Sustainability compliance classification (Compliant, Non-compliant, Partially compliant).

c) Algorithm 3.3: Downstream Sustainability Assessment (Environmental Compliance):

Input: Cleaned text data from ESG reports for downstream

1. Extract sustainability metrics relevant to downstream operations:
   a. Refinery emissions and waste management (UNSDG 13)
   b. Renewable energy adoption in refining (UNSDG 7)
2. Compare extracted metrics with predefined thresholds:
   a. Emission reductions > X% → Compliant
   b. Renewable energy usage > Y→ Compliant
   c. If reductions are between X% and (X-Δ) % for emissions or renewable energy usage is slightly below Y% →Partially compliant
   d. Else → non-compliant
3. Classify environmental compliance.

Output: Sustainability compliance classification (Compliant, Non-compliant, Partially compliant)

D. Algorithm 4: Greenwashing Detection

These algorithms assess greenwashing detection across various segments of OGSC.

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a) Algorithm 4.1: Upstream Greenwashing Detection:

Input: ESG report for upstream company

1. Identify vague or overly positive claims related to:
    a. Emissions reduction (without supporting data)
    b. Water management (without concrete figures)
2. Use sentiment score and keyword matching to flag potential greenwashing:
    a. Positive sentiment with vague language → Flag as greenwashing
    b. Neutral or negative sentiment → Legitimate claims

Output: Greenwashing flags in the report

b) Algorithm 4.2: Midstream Greenwashing Detection:

Input: ESG reports for midstream

1. Identify exaggerated claims related to:
    a. Low emissions in transportation
    b. Use of renewable energy in midstream operations
2. Use sentiment score and keyword analysis to detect unsupported claims:
    a. Positive sentiment without specific metrics → Flag as greenwashing
    b. Neutral or negative sentiment → Legitimate claims

Output: Greenwashing flags in the report

c) Algorithm 4.3: Downstream Greenwashing Detection:

Input: ESG reports for downstream

1. Identify potentially misleading claims related to:
    a. "Eco-friendly" marketing in product labelling
    b. Energy mix claims without quantifiable data
2. Use sentiment analysis and keyword matching to detect unsupported claims:
    a. Positive sentiment without data → Flag as greenwashing
    b. Neutral or negative sentiment → Legitimate claims

Output: Greenwashing flags in the report.

This process has been run for each company based on its profile as upstream, midstream or downstream. Thus, a total of 15 company profiles are built by using this framework to defend the OGSC. All such relevant data and reports are sourced from publicly available company websites and UN sustainability data repositories, which are openly available for all with full accesses of reading and downloads.

V. Data Analysis and Result

The analysis of 15 such prominent OGSC companies (stream-wise) has been performed in Python programming in Google Colab platform. Various python libraries and packages are used to build this framework. Some of the major tools are pdfminer, used to extract raw text from the .pdf extension reports; nltk (Natural Language Toolkit) to pre-process texts by tokenizing, stop-word removal, and lemmatization; re (Regular Expressions) for cleaning special characters, numbers, and unnecessary symbols; TextBlob for sentiment analysis; NumPy, pandas, matplotlib, seaborn and scikit-learn together used for efficient data analysis, visualization and error analysis purposes. Table 3 shows the analyses of the study in details. Predicted labels are marked by using this proposed framework and true labels are the original claims made by the reputed international auditing firms and policy regulators throughout the world on those published reports.

Table 3: Analysis of the Proposed Model

Company	True Label		Predicted Label	
	EC	***GW***	***EC***	***GW***
U1	C	LC	PC	LC
U2	PC	PG	PC	PG
U3	C	PG	C	PG
U4	C	LC	C	PG
U5	PC	LC	PC	LC
M1	C	LC	C	LC
M2	PC	PG	PC	PG
M3	NC	PG	NC	PG
M4	C	LC	C	PG
M5	PC	PG	PC	PG
D1	PC	PG	PC	PG
D2	C	PG	C	PG
D3	C	LC	PC	LC
D4	PC	LC	PC	LC
D5	C	LC	C	LC

EC: Environmental Compliance; GW: Greenwashing; U, M, D: Upstream, Midstream and Downstream Companies from serial numbers 1 to 5; C, PC, NC: Compliant, Partially Compliant and Non-compliant; LC, PG: Legitimate Claims and Possible Greenwashing, respectively.

The performance of proposed framework has been examined thoroughly under the metrics of accuracy, precision, recall, F1 score, Receiver Operating Characteristic Area Under the Curve (ROC AUC) score, and confusion matrix. Table 4 shows such values separately for Environmental Compliance (EC) and Greenwashing (GW) at different supply chain stages as well as for the complete OGSC.

Table 4: Performance Metrics Streamwise and Feature wise

Performance Metrics	Upstream		Midstream		Downstream		OGSC	
	EC	GW	EC	GW	EC	GW	EC	GW
Accuracy	0.80	0.80	1	0.80	0.80	1	0.87	0.87
Precision	0.83	0.83	1	0.88	0.83	1	0.92	0.89
Recall	0.83	0.83	1	0.75	0.83	1	0.92	0.88
F1 Score	0.80	0.80	1	0.76	0.80	1	0.90	0.87

Above result highlights overall robustness of the framework, it has the best performance in predicting EC values for midstream and GW at downstream followed by EC, GW of the OGSC; EC of downstream; and EC, GW of the upstream, respectively. Figure 2 represents comparison of such performance metrics across different supply chain stages by combining EC and GW together to showcase overall robustness of the proposed framework.

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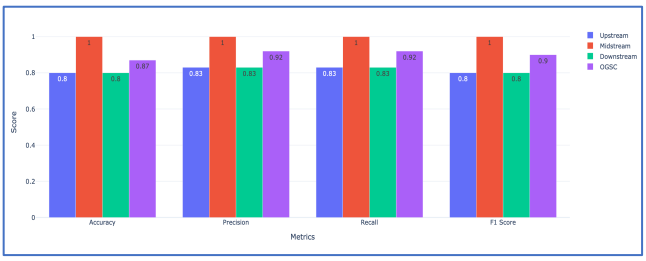

Figure 2: Performance Metrics Comparison across Upstream, Midstream, Downstream and Combined OGSC

In the upstream sector, while the model performs well, improvement in addressing partially compliant cases required. Future iterations should focus on refining feature extraction from ESG reports while keeping a balance of benchmarking with UNSDG targets. Midstream segment achieves perfect scores in compliance and greenwashing detection, likely due to less ambiguous ESG report claims and indicates a well-suited sentiment analysis model for this stage. Downstream shows strong compliance detection, with perfect precision in greenwashing detection, thus the keyword extractions and sentiment classification together effectively capture the ESG claims.

There are a total of 15 companies involved 5 each in 3 segments and 2 types of features extractions for them are performed, thus a pool of 30 observations have been registered. The ROC AUC score is 0.88 for this study. Figures 3 and 4 show the performance and confusion matrix of the proposed framework, respectively.

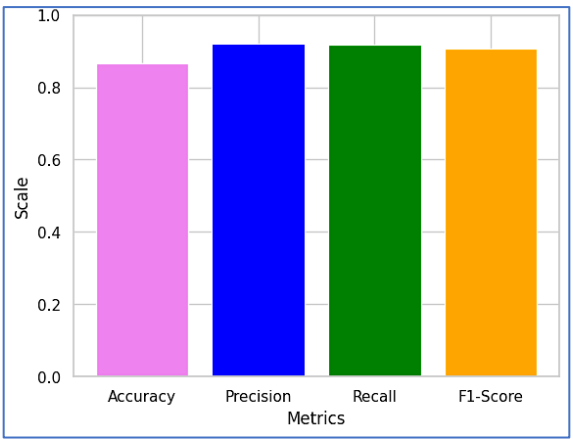

Figure 3: Overall Performance of the Framework

Figure 4: Confusion Matrix of the Framework

Errors in the model reveals that misclassifications mainly occur in partially compliant cases, where the framework struggles to differentiate between genuine sustainability claims and greenwashing. Confusion matrices and ROC curve analysis indicate that while the framework is generally effective, further enhancements in feature extraction and sentiment analysis are required to improve the accuracy of the model.

In summary, the framework demonstrates robustness across the OGSC, however performance slightly deteriorates while aggregating results due to the complexities involved in a global supply chain. Upstream and downstream operations often involve in subjective claims at ESG reporting, these might be due to various complex network of stages involved in such operations. Thus, it is challenging for NLP based model to perfectly analyze and interpret such streams, while midstream data is more structured due to lesser complexities in the operations and/ or might be using a set structure of ESG reporting practices worldwide.

VI. Conclusion

The novelty of this framework lies within the dual focus on analyzing sustainability efforts and greenwashing detection together using NLP techniques. Complex textual data of ESG reports have been benchmarked with the UNSDG targets and thus evaluated to build a global framework. Unlike existing models, this framework unifies reporting transparency along with scalability. It offers a data driven methodology for academic discourse in OGSC, allows self-assessment of sustainability efforts against global benchmarks and identifies misleading claims for businesses. For regulators and policy makers, this study provides a tool for assessing such corporate claims and monitoring the progress to achieve UNSDG targets. And for the society, it promotes accountability and transparency in the OGSC by safeguarding public trust and informed decision-making. Despite such contributions, there are some limitations such as over reliance on ESG reports for the data analysis, sample size of the study, keyword matching, etc., which demand for some further studies on adaptability and accuracy of the model using more datasets, collaborative work with businesses and regulatory bodies to improve keyword searches. In summary, this framework might help to build a global framework to assess the genuine efforts adopted by the businesses towards global sustainability by utilizing ML approaches.

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